Evaluation 2 Report – Interactive Prototype 2

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1. Introduction

This report presents the results and analysis of user testing for Interactive Prototype 2 of *Procreate XR*, an immersive drawing and creation tool designed for mixed reality. The main goal of this evaluation was to test how users interact with the core XR tools and to collect feedback on usability, comfort, and satisfaction.

Five participants were invited to complete five structured tasks:

1. Drawing with the 3D pen
2. Using two eraser effects
3. Pressing the physical reset button
4. Opening the wrist menu
5. Remotely grabbing and floating tools

Each task measured completion time, success rate, accuracy, satisfaction, and user comments.

2. Summary of Findings

2.1 Overall Performance

All participants completed every task successfully, demonstrating that the prototype was functional and stable. The average completion time for each task ranged from 11 to 33 seconds, and no major usability breakdowns occurred.

In general, participants found the drawing, erasing, and remote grabbing experiences smooth and intuitive. However, the physical reset button and wrist menu lacked polish or functionality, which limited the overall interaction depth.

2.2 Task Analysis

Task	Key Findings
1. Drawing with the 3D Pen	All five participants completed the task easily. Drawing felt natural, with average satisfaction 4.2/5. One issue occurred when

	the pen penetrated the canvas, suggesting collision detection needs refinement.
2. Erasing with Two Eraser Modes	Users appreciated the soft eraser's realism. The hard eraser felt less intuitive for some. Average satisfaction 4.4/5.
3. Physical Reset Button	Although the button functioned correctly, users reported mechanical instability and inconsistent sound. Some needed extra force to press. Average satisfaction 3.2/5.
4. Wrist Menu Access	All participants opened the menu successfully, but it contained no working tools. Users rated it easy to access but not useful. Satisfaction 3.0/5.
5. Remote Grab and Floating Tools	The strongest feature. All participants found it realistic, convenient, and efficient. Accuracy 5/5 and satisfaction 5/5.

2.3 Quantitative Summary

Metric	Average Result	Observation
Completion Time	~22 seconds per task	Acceptable for early prototype
Success Rate	100%	All tasks completed
Accuracy (1–5)	4.3	Good control performance
Satisfaction (1–5)	3.9	High enjoyment overall
Common Issues	Missing sound, canvas collision, button instability	Mostly mechanical and polish-related

3. User Feedback Overview

Participants' comments were mostly positive toward the natural interaction design. Common phrases included *"smooth," "realistic,"* and *"fun."*

- Positive Themes: Smooth drawing, convenient remote grab, intuitive controls.
- Negative Themes: Button stiffness, missing audio cues, inactive wrist menu.

These insights indicate that the physical interaction concept works well but needs refinement for realism and consistency.

4. Discussion

Prototype 2 successfully validated several key interaction concepts. The 3D pen and eraser offered realistic spatial drawing experiences, and the remote grab feature provided a strong sense of convenience and immersion. These features demonstrate the potential of spatial creation workflows in XR.

However, the physical reset button revealed mechanical design challenges, including detachment and inconsistent feedback. The wrist menu, while accessible, lacked working tools, limiting user exploration. These issues suggest that functional completeness and feedback systems (sound, haptics, and visual cues) are essential for improving user experience.

Additionally, the minor canvas penetration issue shows the importance of precise collision handling between pen tips and canvas surfaces. Addressing this will increase realism and prevent visual confusion.

5. Iteration Plan

Based on the evaluation, the next iteration (Prototype 3) will focus on the following improvements:

5.1 Enhancing Drawing and Erasing

- Add pen stroke sound effects and haptic feedback when contacting the canvas.
- Improve collision detection to prevent pen penetration.
- Optimize eraser response time for smoother control.

5.2 Improving Physical Buttons

- Refine button alignment to avoid detachment between parts.
- Ensure consistent sound feedback when pressed.
- Assign functional mappings (e.g., toggle passthrough, scene switch, home return).

5.3 Expanding Wrist Menu

- Implement real tool functions (pen, eraser, reset, color picker).
- Add customization options such as brush size and opacity.
- Provide clearer visual feedback when the menu opens.

5.4 System and Onboarding

- Introduce a main menu/home scene for navigation.
- Include tutorial instructions explaining gestures and tools.
- Add visual indicators for interactive objects (e.g., “press here”).

6. Conclusion

This evaluation demonstrated that Prototype 2 provided an enjoyable and functional XR drawing experience. All participants successfully completed their tasks, showing that the interaction system was stable and intuitive.

The strongest feature was the remote grab and floating tool system, which significantly improved usability. The weakest elements were the reset button and wrist menu, both requiring functional and feedback enhancements.

By addressing these issues, Prototype 3 aims to achieve a more polished, realistic, and interactive creative environment, aligning closely with the project's goal of an immersive Procreate-style experience in XR.